

SUMMARY TABLE: PHYSICS GRADE 10

Sub-Strand 2.1: Properties of Waves — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Properties of Waves
Driving Question	DRIVING QUESTION: How can we hear a matatu around a corner that we cannot see, and why does its pitch change and its radio signal fade as it moves?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Matatu Mystery: Launching the Phenomenon	Write what you observed about the three puzzles: the music heard around the corner while the torch beam was blocked by the wall, the matatu horn (true frequency 400 Hz) heard at 418 Hz while approaching and 383 Hz while receding at 15 m/s, and the 100 MHz FM radio fading at certain spots near the Ngong Hills though the station never changes.	Write what you learned today: a wave is a disturbance that carries energy from one place to another without carrying matter (as shown by the slinky pulse and the sound from the matatu horn), and good scientific questions are testable.	Write what you can explain so far, and what you cannot yet explain: give your best first idea for each of Parts A (corner), B (pitch) and C (fading signal) in your model, and list one question from the Driving Question Board that you want answered next.
Lesson 2 Waves Travel in Straight Lines: Rays, Wavefronts and the Blocked Torch Beam	In the ripple tank at 10 Hz in deep water we saw straight, evenly spaced plane wavefronts with wavelength 3.0 cm, and we calculated $v = 0.30$ m/s. A torch beam through a slit travelled in a straight line, and a block in its path made a dark shadow with a sharp edge.	In a uniform medium waves travel in straight lines (rectilinear propagation). A wavefront joins points in phase, and a ray is drawn at 90 degrees to the wavefronts to show direction. Light travels at 3.0×10^8 m/s in air and obeys this rule.	We can explain why a torch beam from the matatu is blocked by the concrete wall: the light rays travel straight and cannot pass through or around the wall, so a shadow forms where Wanjiru and Otieno stand. We cannot yet explain why the music is heard around the corner, and that question stays on the Driving Question Board.
Lesson 3 Bouncing Back: Reflection and the Laws of Reflection	We observed ripples (10 Hz, wavelength 3.0 cm) bouncing off a barrier, light rays bouncing off a plane mirror with the angle of reflection equal to the angle of incidence, and an echo returning after 0.50 s from a wall 85 m away.	Waves reflect from surfaces according to the law of reflection: the angle of incidence i equals the angle of reflection r , measured from the normal, and the incident ray, reflected ray and normal lie in the same plane. Reflection does not change the frequency, wavelength or speed of the	Sound from the matatu can reach Wanjiru and Otieno by reflecting from walls, and the Ngong Hills can reflect the FM radio signal, giving a second signal that will later be used to explain the fading spots. Reflection alone does not yet explain how sound bends around the corner (Part A), why the pitch

		wave in the same medium. An echo is a reflected sound wave.	changes (Part B), or why the signal fades (Part C), so we need more evidence in later lessons.
Lesson 4 Echoes, Sonar and Hall Design: Putting Reflection to Work	We observed that a clap 85 m from a wall returns as an echo after 0.50 s, that a closer wall gives a delay too short to hear separately, and that a Lake Victoria sonar pulse returned after 0.040 s.	We learned that sound reflects from hard surfaces following the law of reflection, that $d = vt/2$ because the sound travels there and back, that a separate echo needs at least 17 m in air, and that hard and soft surfaces change how a hall sounds.	We can explain that the sonar depth is $1500 \times 0.040 / 2 = 30$ m and that reflection can send some matatu sound along the street, but reflection alone does not explain why sound reaches people around the 4.0 m opening, why the horn pitch changes, or why the FM signal fades.
Lesson 5 Refraction: Speed and Direction Change	In the ripple tank, waves moved from deep water ($v = 0.30$ m/s, wavelength 3.0 cm) into shallow water ($v = 0.20$ m/s, wavelength 2.0 cm) at 10 Hz, and the wavefronts bunched closer and the direction turned toward the normal. In the PhET simulation a light ray in air at 40 degrees entered water and bent to about 29 degrees, toward the normal; going from water to air it bent away.	Refraction is a change of direction of a wave caused by a change of wave speed at a boundary. A wave that slows down bends toward the normal; a wave that speeds up bends away from the normal. The frequency stays the same while the speed and wavelength change.	Refraction describes how waves behave when they enter a new medium or depth, but it does not explain why the matatu music is heard around the building, because the sound stays in the same air. We still need other ideas to explain the corner, the pitch change and the fading radio signal near the Ngong Hills.
Lesson 6 Refraction Calculations and the Bent Straw: Why Wavelength Changes but Frequency Does Not	In the ripple tank at 10 Hz, the waves had a wavelength of 3.0 cm in deep water and 2.0 cm in shallow water and bent at the boundary. A ray of light entering water at 40 degrees bent to about 29 degrees from the normal. A straw in water looked broken at the surface.	When a wave is refracted, its frequency stays the same because the source sets it, while its speed and wavelength change ($v = f\lambda$). The refractive index $n = \sin i / \sin r = c/v$; for water $n = 1.33$ and light travels at 2.26×10^8 m/s. Wavelength depends on the speed of the wave and the frequency of the source.	The straw looks bent because light from its underwater part bends away from the normal as it leaves the water, and our eyes trace the rays back in straight lines to a shallower apparent position. The straw itself is straight. This does not yet explain why sound turns the corner, but we now have the tool $v = f\lambda$ to work out wavelengths, such as 0.85 m for the matatu horn at 400 Hz, which we will use in Lessons 7 and 8.
Lesson 7 Diffraction: Gap Size and Wavelength	In the ripple tank at 10 Hz ($\lambda = 3.0$ cm), waves passing through a wide gap of 12 cm mostly travelled straight, while waves passing through a narrow gap of 3 cm or 1.5 cm spread out in almost circular wavefronts. The PhET simulation showed the same pattern when slit width or wavelength was changed. The street opening around the building is 4.0 m wide.	Diffraction is the spreading of waves around an obstacle or through a gap. It is greatest when the wavelength is comparable to or bigger than the gap, and is very small when the wavelength is much smaller than the gap. The wavelength can be found from $v = f\lambda$, for example $0.30 \div 10 = 0.03$ m = 3.0 cm.	This partly explains Part A of the matatu puzzle: waves can bend around the corner if their wavelength is comparable to the 4.0 m opening. We still need to compare the wavelengths of sound and light with 4.0 m in Lesson 8 to explain why the music is heard around the corner but the torch beam is blocked.
Lesson 8 Diffraction and the Corner Puzzle: Why Sound Turns the Corner	We observed that a low tone (about 200 Hz) was heard more clearly than a high tone (about 1000 Hz) around the corner of the partition, while the torch beam did not	We learned that diffraction is greatest when the wavelength is comparable to or bigger than the gap. For the horn, the ratio of wavelength to gap is $0.85/4.0 = 0.21$, so the	Wanjiru and Otieno hear the matatu music around the building because the sound waves of wavelength 0.34 m to 1.7 m are comparable to the 4.0 m opening and

but Light Does Not	reach us. We calculated that the matatu horn (400 Hz) has wavelength 0.85 m, the music ranges from 1.7 m (200 Hz) to 0.34 m (1000 Hz), and light has wavelength 5.0×10^{-7} m, compared with the 4.0 m street opening.	sound clearly bends, while for light the ratio is about 1.3×10^{-7} , so it hardly bends at all. Lower pitches (longer wavelengths) bend around the corner more than higher pitches.	diffract around the corner, while the torch beam, with wavelength 5.0×10^{-7} m, hardly diffracts and is blocked by the wall. The pitch change and the fading radio signal are still open questions on our Driving Question Board.
Lesson 9 Interference and the Fading Radio: Loud Spots, Quiet Spots and Weak FM Signal	In the ripple tank, two sources made bands of strong ripples and calm water. With two speakers playing 340 Hz we found loud and quiet spots in the room, and the quiet spots vanished when one speaker was covered.	Overlapping waves add together (superposition). When the path difference is $n\lambda$ the waves reinforce (constructive interference, loud or strong). When it is $(n+1/2)\lambda$ they cancel (destructive interference, quiet or weak). For 340 Hz, $\lambda = 1.00$ m, so a path difference of 1.00 m gives a loud spot and 0.50 m gives a quiet spot.	Near the Ngong Hills, the 100 MHz FM signal has $\lambda = 3.0$ m. The direct signal and the signal reflected from the hillside overlap and cancel at weak spots about 1.5 m apart, then reinforce between them. The radio fades and returns even though the station never changes.
Lesson 10 Standing Waves and Musical Notes: Guitar Strings, Nodes and the Half-Wavelength Link	We observed that a slinky and a guitar string of length 0.65 m, when shaken or plucked, showed fixed points of no movement (nodes) and points of large movement (antinodes). The fundamental pattern had 2 nodes and 1 antinode; the second pattern had 3 nodes and 2 antinodes. A shorter string (0.325 m) and a tighter string gave a higher note.	We learned that a standing wave forms when a wave and its reflection travel in opposite directions and superpose. Adjacent nodes are $\lambda/2$ apart, and for the fundamental $L = \lambda/2$. With $v = 143$ m/s on a 0.65 m string, $\lambda = 1.30$ m and $f = 110$ Hz; the second harmonic is 220 Hz, and a 0.325 m string also gives 220 Hz. Length and tension set the note.	We explained that the weak FM spots near the Ngong Hills are about 1.5 m apart because the direct and reflected signals overlap, and the 100 MHz signal has wavelength 3.0 m, so weak spots are half a wavelength (1.5 m) apart, just as nodes on a string are $\lambda/2$ apart. This strengthens Part C of our model, while the pitch change of the matatu horn (Part B) still needs explaining.
Lesson 11 The Doppler Effect: Why the Matatu Horn Changes Pitch	We observed that the pitch of a moving horn or siren sounds higher when the source moves toward us and lower when it moves away, and that the effect is bigger for faster sources, even though the source itself does not change (matatu horn 400 Hz heard as 418 Hz and 383 Hz; ambulance 700 Hz heard as 744 Hz and 661 Hz).	We learned that the Doppler effect is the apparent change in frequency because of relative motion between source and observer: approaching $f' = f v/(v - v_s)$ and receding $f' = f v/(v + v_s)$, with $v = 340$ m/s for sound and wavelength and frequency linked by the wave equation $v = f\lambda$.	We explained that Baraka's horn stays at 400 Hz, but the moving matatu (15 m/s) compresses the waves ahead of it so Otieno hears 418 Hz, and stretches them behind it so Wanjiru hears 383 Hz, a drop of 35 Hz. This completes Part B of our class model; the corner puzzle (Part A) and the fading radio (Part C) have different causes.
Lesson 12 Applications and Synthesis: FM, Ultrasound and the Final Explanation of the Matatu Phenomenon	We observed that music is heard around the building but the torch beam is blocked; the matatu horn sounds 418 Hz approaching and 383 Hz receding; the FM radio fades near the Ngong Hills about every 1.5 m; the FM carrier is 100 MHz and changes frequency with a 1 kHz tone (up to 75 kHz) while amplitude stays constant; ultrasound of 3.5 MHz in soft tissue at 1540 m/s has a wavelength of 0.44 mm and its echoes make an image.	We learned that sound waves of wavelength 0.34 m to 1.7 m are comparable to the 4.0 m opening and diffract, while light of 5.0×10^{-7} m hardly does; that the Doppler effect changes the frequency heard when a source moves; that interference of direct and reflected signals makes weak and strong spots; and that FM radio and ultrasound imaging apply frequency, wavelength, reflection and the wave equation $v = f\lambda$.	The music is heard around the corner because sound diffracts around the 4.0 m opening (0.85 m wavelength, ratio 0.21) but light does not. The horn really stays at 400 Hz; it sounds 418 Hz approaching and 383 Hz receding because the moving matatu compresses waves ahead and stretches them behind (Doppler effect). The FM signal fades near the Ngong Hills because the direct and reflected 3.0 m waves interfere destructively about 1.5 m apart, while the station never changes.

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